

ISIT312 Big Data Management

SQL for Data Warehousing

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SQL for Data Warehousing

Outline

SQL/OLAP Operations

Window partitioning

Window ordering

Window framing

SQL/OLAP Operations

Consider the **SALES** fact table

To compute all possible aggregations along the dimensions **Product** and **Customer** we must scan the whole relational table **SALES** several times

It can be implemented in SQL using **NULL** and **UNION** in the following way:

Finding aggregations along many dimensions

```
SELECT ProductKey, CustomerKey, SalesAmount
FROM Sales
UNION
SELECT ProductKey, NULL, SUM(SalesAmount)
FROM Sales
GROUP BY ProductKey
UNION
SELECT NULL, CustomerKey, SUM(SalesAmount)
FROM Sales
GROUP BY CustomerKey
UNION
SELECT NULL, NULL, SUM(SalesAmount)
FROM Sales;
```

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SQL/OLAP Operations

A data cube created through **UNION** of individual **SELECT** statements each one creating one combination of dimensions looks in the following way

Data cube

ProductKey	CustomerKey	SalesAmount
p1	c1	100
p2	c1	70
p3	c1	30
NULL	c1	200
p1	c2	105
p2	c2	60
p3	c2	40
NULL	c2	205
p1	c3	100
p2	c3	40
p3	c3	50
NULL	c3	190
p1	NULL	305
p2	NULL	170
p3	NULL	120
NULL	NULL	595

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SQL/OLAP Operations

Computing a cube with n dimensions requires $(2*2*2*... *2)$ (n times) **SELECT** statements with **GROUP BY** clause

SQL/OLAP extends the **GROUP BY** clause with the **ROLLUP** and **CUBE** operators

ROLLUP computes group subtotals in the order given by a list of attributes

CUBE computes all totals of such a list

Shorthands for a more powerful operator, **GROUPING SETS**

Equivalent queries

Sample application of ROLLUP operation

```
SELECT ProductKey, CustomerKey, SUM(SalesAmount)
FROM Sales
GROUP BY ROLLUP(ProductKey, CustomerKey);
```

Sample application of GROUPING SET operation

```
SELECT ProductKey, CustomerKey, SUM(SalesAmount)
FROM Sales
GROUP BY GROUPING SETS((ProductKey, CustomerKey), (ProductKey), ());
```

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SQL/OLAP Operations

Equivalent queries

Sample application of CUBE operation

```
SELECT ProductKey, CustomerKey, SUM(SalesAmount)
FROM Sales
GROUP BY CUBE(ProductKey, CustomerKey);
```

Sample application of GROUPING SETS operation

```
SELECT ProductKey, CustomerKey, SUM(SalesAmount)
FROM Sales
GROUP BY GROUPING SETS((ProductKey, CustomerKey), (ProductKey), (CustomerKey), ());
```

SQL/OLAP Operations

GROUP BY ROLLUP

ProductKey	CustomerKey	SalesAmount
p1	c1	100
p1	c2	105
p1	c3	100
p1	NULL	305
p2	c1	70
p2	c2	60
p2	c3	40
p2	NULL	170
p3	c1	30
p3	c2	40
p3	c3	50
p3	NULL	120
NULL	NULL	595

GROUP BY CUBE

ProductKey	CustomerKey	SalesAmount
p1	c1	100
p2	c1	70
p3	c1	30
NULL	c1	200
p1	c2	105
p2	c2	60
p3	c2	40
NULL	c2	205
p1	c3	100
p2	c3	40
p3	c3	50
NULL	c3	190
NULL	NULL	595
p1	NULL	305
p2	NULL	170
p3	NULL	120

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Window partitioning

Allows to compare detailed data with aggregate values

For example, find a relevance of each customer with respect to the sales of the product

Sample window partitioning

```
SELECT ProductKey, CustomerKey, SalesAmount,  
       MAX(SalesAmount) OVER (PARTITION BY ProductKey) AS MaxAmount  
FROM SALES;
```

First three columns are obtained from the **SALES** table

The fourth column is created in the following way

- Create a window called **partition** that contains all tuples of the same product
- **SalesAmount** is aggregated over this window using **MAX** function

Window partitioning

Sample window partitioning

```
SELECT ProductKey, CustomerKey, SalesAmount,  
       MAX(SalesAmount) OVER (PARTITION BY ProductKey) AS MaxAmount  
FROM SALES;
```

ProductKey	CustomerKey	SalesAmount	MaxAmount
p1	c1	100	105
p1	c2	105	105
p1	c3	100	105
p2	c1	70	70
p2	c2	60	70
p2	c3	40	70
p3	c1	30	50
p3	c2	40	50
p3	c3	50	50

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Window ordering

ORDER BY clause allows the rows within a partition to be ordered

It is useful to compute rankings, with a function **RANK ()**

For example, how does each product rank in the sales of each customer

Sample window ordering

```
SELECT ProductKey, CustomerKey, SalesAmount,  
       RANK() OVER (PARTITION BY CustomerKey ORDER BY SalesAmount DESC) AS RowNo  
FROM Sales;
```

Product Key	Customer Key	Sales Amount	RowNo
p1	c1	100	1
p2	c1	70	2
p3	c1	30	3
p1	c2	105	1
p2	c2	60	2
p3	c2	40	3
p1	c3	100	1
p3	c3	50	2
p2	c3	40	3

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Window framing

It is possible to define a size of a partition

It can be used to compute statistical functions over time series, like moving average

For example, three-month moving average of sales by product

```
SELECT ProductKey, Year, Month, SalesAmount,  
       AVG(SalesAmount) OVER (PARTITION BY ProductKey  
                              ORDER BY Year, Month  
                              ROWS 2 PRECEDING) AS MovAvg  
FROM SALES;
```

Sample window framing

Processing of a query opens a window with the rows pertaining to the current product

Then, it orders the window by year and month and computes the average over the current row and the previous two ones if they exist

Window framing

Sample window framing

```
SELECT ProductKey, Year, Month, SalesAmount,
       AVG(SalesAmount) OVER (PARTITION BY ProductKey
                              ORDER BY Year, Month
                              ROWS 2 PRECEDING) AS MovAvg
FROM SALES;
```

Product Key	Year	Month	Sales Amount	MovAvg
p1	2011	10	100	100
p1	2011	11	105	102.5
p1	2011	12	100	101.67
p2	2011	12	60	60
p2	2012	1	40	50
p2	2012	2	70	56.67
p3	2012	1	30	30
p3	2012	2	50	40
p3	2012	3	40	40

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Window framing

Another example, a year-to-date sum of sales by product

```
SELECT ProductKey, Year, Month, SalesAmount,  
       SUM(SalesAmount) OVER (PARTITION BY ProductKey, Year  
                              ORDER BY Month  
                              ROWS UNBOUNDED PRECEDING) AS YTD  
FROM SALES;
```

Sample window framing

Processing of a query, opens a window with the tuples of the current product and year ordered by month

AVG () is applied to all rows before the current row (**ROWS UNBOUNDED PRECEDING**)

Window framing

Sample window framing

```
SELECT ProductKey, Year, Month, SalesAmount,  
       SUM(SalesAmount) OVER (PARTITION BY ProductKey, Year  
                              ORDER BY Month  
                              ROWS UNBOUNDED PRECEDING) AS YTD  
FROM SALES;
```

Product Key	Year	Month	Sales Amount	YTD
p1	2011	10	100	100
p1	2011	11	105	205
p1	2011	12	100	305
p2	2011	12	60	60
p2	2012	1	40	40
p2	2012	2	70	110
p3	2012	1	30	30
p3	2012	2	50	80
p3	2012	3	40	120

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References

A. VAISMAN, E. ZIMANYI, Data Warehouse Systems: Design and Implementation, Chapter 5 Logical Data Warehouse Design, Springer Verlag, 2014